Keshav Sharma

India | 8168289407 | keshavsharma71982@gmail.com | LinkedIn Link | GitHub Link | Portfolio

EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, C/C++, SQL, JavaScript

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Multithreading, TCP/IP, REST APIs, System Design, Git/GitHub

Machine Learning & Data Science: Scikit-learn, PyTorch, Pandas, NumPy, Feature Engineering, Statistical Analysis, Anomaly Detection, Time-Series Analysis, Model Evaluation

Backend & Full Stack: FastAPI, Flask, React.js, Streamlit, SQLAlchemy

Databases & Systems: SQLite, MySQL, MongoDB, Redis, In-Memory Systems, Caching, WAL/Persistence

Data Analytics & Visualization: Power BI, DAX, Matplotlib, Plotly, Exploratory Data Analysis

Engineering Practices: PyTest, Test-Driven Development, Debugging, Data Preprocessing, Reproducible Experimentation

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship organized by 1M1B covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value database in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, O(1) LRU caching, and TTL expiration for concurrent request processing.
- Implemented Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery to ensure durable persistence and data consistency under failures and concurrent workloads.
- Developed 46 PyTest tests covering concurrency, edge cases, and recovery scenarios; benchmarked 619K SET ops/sec in-memory and 2.8K ops/sec with WAL persistence to validate reliability and performance.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end financial analytics pipeline processing 50K+ PaySim transactions, combining statistical outlier analysis, duplicate detection, vendor-velocity checks, HHI concentration analysis, and Isolation Forest for transaction-risk detection.
- Designed a star-schema SQLite data warehouse and composite 0–100 risk scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Built a 5-page Power BI audit dashboard with 20+ DAX measures to surface vendor risk, anomaly rates, transaction trends, and risk insights, supported by automated pipeline tests.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack Python application using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP to automate receipt processing and reduce processing time by 40%.
- Engineered and validated REST APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export for reliable application workflows.
- Deployed the application on Render and Vercel, implementing end-to-end workflows across AI processing, backend services, database operations, and frontend validation.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** — **2026**